FinTrust Business & Data Intelligence Assessment

Summary Documentation

1. Business Understanding

The purpose of this analysis is to understand FinTrust's customer base and transaction activity and identify how data can support business decision-making.

The key business questions include:

- **Which customer segments generate the most transaction activity and value?**
- **Which transaction types and channels are most frequently used?**
- **What is the overall transaction success rate?**
- **What proportion of transactions are flagged for risk review?**
- **Are there differences in transaction behaviour across customer segments?**
- **How does digital engagement relate to customer activity?**
- **Which areas may require further investigation or improvement?**

The analysis can support decisions relating to customer engagement, digital banking channels, transaction performance, risk monitoring and business performance.

Key stakeholders who could benefit include senior management, finance, operations, risk and compliance, customer service, marketing, digital banking and Business Intelligence teams.

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2. Data Understanding and Quality

The analysis used two datasets: Customer and Transaction.

The Customer dataset contains 1,500 records and 12 columns, including customer demographics, customer segment, account information and digital engagement.

The Transaction dataset contains 12,000 records and 11 columns, including transaction date/time, transaction type, transaction amount, channel, transaction status and risk-review information.

The datasets are connected using Customer_ID, allowing customer characteristics to be analysed alongside transaction behaviour.

Initial data profiling identified no missing values in the Customer dataset. The Transaction dataset contained 192 missing values, consisting mainly of 96 missing Device_Type values and 96 missing Location values. These categorical values can be represented as "Unknown" where appropriate rather than removing the affected transaction records.

No duplicate records were identified during the initial profiling. Potential outliers were identified in transaction amounts and should be investigated further rather than automatically removed, as high-value transactions may represent legitimate customer activity.

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3. Key Analysis and Findings

SQL and Python were used to analyse transaction activity, customer segments, transaction channels, transaction status and risk-review patterns.

Customer Segments

Everyday customers are the largest customer segment, with 711 customers (47.4%).

When transaction activity was analysed by customer segment, Everyday customers generated the highest transaction volume with 5,644 transactions and approximately ₦261.46 million in total transaction value.

SME customers recorded the highest average transaction value at approximately ₦49,115.69 per transaction.

This shows that transaction volume and average transaction value provide different perspectives when evaluating customer segments.

Transaction Activity

Transfers recorded the highest transaction volume with 3,549 transactions, followed by Card Purchases with 3,033 transactions.

The Mobile App was the most frequently used transaction channel, recording 5,102 transactions, approximately 42.5% of all transactions, with approximately ₦240.1 million in transaction value.

Transaction Performance

The majority of transactions were successful:

- **Successful – 90.5%**
- **Failed – 5.2%**
- **Reversed – 2.7%**
- **Pending – 1.6%**

Although most transactions were successful, the remaining transactions provide an opportunity for further analysis, particularly by channel and transaction type.

Risk Review

A total of 2,352 transactions (19.6%) were flagged for risk review, representing approximately ₦161.78 million in transaction value.

Further analysis should examine whether risk-review activity is concentrated within particular transaction types, channels, customer segments or international transactions.

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4. KPI and Dashboard Plan

The proposed Power BI dashboard will provide management with a clear view of customer and transaction performance.

Key KPIs will include:

- **Total Customers**
- **Total Transactions**
- **Total Transaction Value**
- **Average Transaction Value**
- **Transaction Success Rate**
- **Risk Review Rate**
- **International Transaction Rate**
- **Average Digital Engagement Score**

The dashboard will include transaction trends over time, transaction activity by type and channel, transaction status, customer segment performance and risk-review analysis.

Filters will allow users to explore the data by date, customer segment, account type, channel, transaction type, transaction status, international transaction and risk-review status.

Conclusion

The initial analysis demonstrates that FinTrust's customer and transaction data can provide useful business intelligence. The findings highlight differences in customer segment behaviour, transaction channels, transaction performance and risk-review activity.

The next stage of analysis should focus on deeper investigation of transaction failures, risk patterns, customer behaviour and channel performance before developing the final interactive Power BI dashboard.